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Utility vehicle

Abstract

A utility vehicle includes a vehicle body frame, a traveling drive source, a mount bracket for mounting the traveling drive source on the vehicle body frame, and a related component connected to the traveling drive source. The related component is supported by the mount bracket.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7650959	12/2009	Kato	180/311	B60G 7/02
8875833	12/2013	Kouma et al.	N/A	N/A
2004/0154851	12/2003	Massicotte	180/190	B62M 27/02
2015/0086145	12/2014	Kim	384/441	F16C 35/02
2017/0182877	12/2016	Gillespie	N/A	B60K 5/1283
2019/0023115	12/2018	Mackenzie	N/A	B60K 6/387

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
4221694	12/2008	JP	N/A

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Background/Summary

BACKGROUND OF THE INVENTION

Technical Field

(1) The present disclosure relates to a utility vehicle.

Related Art

(2) As described in U.S. Pat. No. 8,875,833, a utility vehicle includes an engine arranged at a vehicle body rear portion and a transmission arranged behind the engine. In the utility vehicle, power from the engine is transmitted to a rear wheel through the transmission, and is transmitted to a front wheel through a propeller shaft arranged on the vehicle body right side of the engine.

SUMMARY

(3) A utility vehicle according to one aspect of the present disclosure includes a vehicle body frame, a traveling drive source, a mount bracket for mounting the traveling drive source on the vehicle body frame, and a related component connected to the traveling drive source. The related component is supported by the mount bracket.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing and the other features of the present invention will become apparent from the following description and drawings of an illustrative embodiment of the invention in which:

(2) FIG. 1 is a schematic configuration diagram of a utility vehicle;

(3) FIG. 2 is a schematic side view of the utility vehicle;

- (4) FIG. 3 is a perspective view illustrating a vehicle body frame and a power unit of a vehicle body rear portion;
- (5) FIG. 4 is a perspective view illustrating a vehicle body frame of a vehicle body rear portion;
- (6) FIG. 5 is a left side view of the power unit;
- (7) FIG. 6 is a right side view of the power unit;
- (8) FIG. 7 is a top view of the power unit;
- (9) FIG. 8 is a front view of the power unit;
- (10) FIG. 9 is a front view of the power unit excluding a first mount bracket;
- (11) FIG. 10 is a perspective view illustrating the vicinity of the first mount bracket;
- (12) FIG. 11 is a perspective view illustrating the vicinity of the first mount bracket excluding the vehicle body frame;
- (13) FIG. 12 is a cross-sectional view of the vicinity of the first mount bracket taken along line Y12-Y12 in FIG. 11;
- (14) FIG. 13 is a cross-sectional view of the vicinity of the first mount bracket taken along line Y13-Y13 in FIG. 11;
- (15) FIG. 14 is a perspective view illustrating the vicinity of a second mount bracket;
- (16) FIG. 15 is a perspective view illustrating the vicinity of the second mount bracket excluding the vehicle body frame;
- (17) FIG. 16 is a perspective view illustrating the second mount bracket;
- (18) FIG. 17 is a perspective view illustrating the vicinity of a rear mount;
- (19) FIG. 18 is a top view of a crankcase and an oil pan;
- (20) FIG. 19 is a cross-sectional view of the crankcase and the oil pan taken along line Y19-Y19 in FIG. 18; and
- (21) FIG. 20 is a cross-sectional view of the crankcase and the oil pan taken along line Y20-Y20 in FIG. 18.

DETAILED DESCRIPTION OF EMBODIMENTS

- (22) Hereinafter, a utility vehicle according to an embodiment of the present disclosure will be described with reference to the accompanying drawings. The utility vehicle is mainly a vehicle for traveling not only on grassland, gravel, and sand, but also on unpaved mountain roads and forest roads, and off-road places, such as muddy and rocky lands. Note that, in the present description, “front”, “rear”, “left”, and “right” refer to a front direction, a rear direction, a left direction, and a right direction as viewed from a driver on the utility vehicle.
- (23) FIG. 1 is a schematic configuration diagram of the utility vehicle, and FIG. 2 is a schematic side view of the utility vehicle. As illustrated in FIGS. 1 and 2, a utility vehicle 1 according to an embodiment of the present disclosure includes a pair of left and right front wheels 2 at a vehicle body front portion and a pair of left and right rear wheels 3 at a vehicle body rear portion. A riding space (vehicle interior) 4 where passengers including a driver get in is provided between the front wheels 2 and the rear wheels 3.
- (24) The riding space 4 is surrounded by a ROPS 5. The ROPS 5 is an abbreviation for a rollover protection structure and is part of a vehicle body frame 6. A seat 7 is arranged in the riding space 4, and an operation unit 8 such as a steering wheel is provided in front of the seat 7.
- (25) A cargo bed 9 is provided behind the riding space 4. The cargo bed 9 is arranged in a vehicle body rear portion, and a power unit 20 including an engine 21 as a traveling drive source and a transmission 31 is arranged below the cargo bed 9. A driving force from the engine 21 is transmitted to the front wheels 2 and the rear wheels 3 via the transmission 31.
- (26) As illustrated in FIG. 5 described later, the engine 21 includes a crankshaft 22 extending in a vehicle width direction, a crankcase 23 supporting the crankshaft 22 from below, a cylinder block 24 attached to an upper surface of the crankcase 23, having a plurality of cylinders 15, and supporting the crankshaft 22 from above, a cylinder head 25 attached to an upper surface of the cylinder block 24, a head cover 26 attached to an upper surface of the cylinder head 25, and an oil

pan 27 attached to a lower surface of the crankcase 23. The engine 21 is a multi-cylinder engine, specifically, a parallel four-cylinder engine in which four of the cylinders 15 are arranged in parallel in the vehicle width direction. In the cylinder block 24, four of the cylinders 15 are formed side by side in the vehicle width direction.

(27) The transmission 31 is arranged behind the engine 21 and is connected to the engine 21 via a connecting bracket 30. Power from the engine 21 is transmitted to the transmission 31 by a belt converter 35 as a power transmission mechanism that transmits power from the engine 21 to the transmission 31. The belt converter 35 is arranged on the vehicle body left side which is one side in the vehicle width direction of the engine 21 and the transmission 31.

(28) The belt converter 35 includes a drive side pulley 36 attached to the crankshaft 22 which is an output shaft of the engine 21, a driven side pulley 37 attached to an input shaft 32 of the transmission 31, and a V belt 38 wound around the drive side pulley 36 and the driven side pulley 37. The belt converter 35 is a V-belt type continuously variable transmission, and is automatically changed in speed in accordance with a rotational speed of the engine 21 or the like.

(29) The transmission 31 includes the input shaft 32 to which power from the engine 21 is transmitted, a front wheel output shaft 33 for transmitting power from the engine 21 to the front wheels 2, and a rear wheel output shaft 34 for transmitting power from the engine 21 to the rear wheels 3. The transmission 31 changes the speed of power transmitted to the input shaft 32 from the engine 21 via a transmission mechanism (not illustrated) and transmits the power to the front wheel output shaft 33 and the rear wheel output shaft 34.

(30) A two-wheel drive four-wheel drive switching device (not illustrated) is attached to the transmission 31. The two-wheel drive four-wheel drive switching device is configured to be switchable between a two wheel drive state in which the input shaft 32 and the rear wheel output shaft 34 are connected and a four wheel drive state in which the input shaft 32, the front wheel output shaft 33, and the rear wheel output shaft 34 are connected.

(31) Power from the engine 21 transmitted to the rear wheel output shaft 34 is transmitted to the left and right rear wheels 3 via a rear wheel drive shaft 10 extending in the vehicle width direction. Power from the engine 21 transmitted to the front wheel output shaft 33 is transmitted to a front wheel propeller shaft 11 extending in a vehicle body front-rear direction. Power from the engine 21 transmitted to the front wheel propeller shaft 11 is transmitted to the left and right front wheels 2 via a front wheel drive shaft 12 extending in the vehicle width direction.

(32) As illustrated in FIG. 1, the front wheel propeller shaft 11 extends from the transmission 31 arranged behind the engine 21 to the vehicle body front side further than the engine 21, and is arranged below the crankcase 23. As illustrated in FIG. 9 to be described later, the crankcase 23 has a recessed portion 41 in which a lower portion of the crankcase 23 is recessed upward and a remaining portion 42 excluding the recessed portion in an axial direction of the crankshaft 22. At least a part of the front wheel propeller shaft 11 is arranged in a space 43 defined below the recessed portion 41 and is located above a lower end portion of the crankcase 23.

(33) The recessed portion 41 provided in a lower portion of the crankcase 23 is formed at a position corresponding to one of four of the cylinders 15, and the remaining portion 42 provided in a lower portion of the crankcase 23 is formed at a position corresponding to the remaining ones of four of the cylinders 15. In the present embodiment, the recessed portion 41 is formed at a position below one of the cylinders 15 on the vehicle body left side, and the remaining portion 42 is formed at a position below three of the cylinders 15 on the vehicle body right side. At least a part of the front wheel propeller shaft 11 is arranged at a position overlapping the crankcase 23 in a vehicle body vertical direction and the vehicle width direction.

(34) FIG. 3 is a perspective view illustrating a vehicle body frame and a power unit of a vehicle body rear portion. FIG. 4 is a perspective view illustrating a vehicle body frame of a vehicle body rear portion. As illustrated in FIG. 3, the power unit 20 including the engine 21 and the transmission 31 is supported by the vehicle body frame 6 constituting a vehicle body rear portion.

(35) As illustrated in FIG. 4, the vehicle body frame 6 includes a pair of left and right main frames 51 extending in the vehicle body front-rear direction, and a pair of left and right side frames 52 arranged outside the left and right main frames 51 in the vehicle width direction and extending in the vehicle body front-rear direction. The main frame 51 has the front side extending linearly and the rear side curved outward in the vehicle width direction toward the rear side. The side frames 52 extend linearly in the vehicle body front-rear direction. The rear sides of the left and right main frames 51 are connected by a floor cross frame 53 extending in the vehicle width direction. The main frame 51 and the side frame 52 support a floor panel.

(36) The vehicle body frame 6 includes a rear frame 54 provided at a rear end portion of the left and right main frames 51. The rear frame 54 includes a pair of left and right lower rear frames 55 extending in the vehicle body front-rear direction from a rear end portion of the left and right main frames 51 to the rear side, a pair of left and right upper rear frames 56 arranged above the left and right lower rear frames 55 and extending in the vehicle body front-rear direction, and left and right front connecting frames 50 respectively connecting front sides of the left and right lower rear frames 55 and upper rear frames 56 and extending in the vehicle body vertical direction.

(37) Each of the left and right lower rear frames 55 and the left and right upper rear frames 56 has the rear side extending linearly, and the front side being curved outward in the vehicle width direction toward the front side. The left and right lower rear frames 55 are connected by first, second, third, and fourth lower cross frames 57, 58, 59, and 60 extending linearly in the vehicle width direction. The first, second, third, and fourth lower cross frames 57, 58, 59, and 60 are arranged apart from each other in the vehicle body front-rear direction.

(38) The vehicle body frame 6 includes a U-shaped frame 61 that is formed in a U-shape extending in the vehicle width direction and protruding downward and connects the rear sides of the left and right upper rear frames 56, and an upper cross frame 62 that linearly extends in the vehicle width direction further on the vehicle body front side than the U-shaped frame 61 and connects the left and right upper rear frames 56.

(39) A pair of left and right rear sub-frames 63 linearly extending to the vehicle body front side is connected to a portion 61a protruding downward of the U-shaped frame 61. The left and right rear sub-frames 63 are arranged between the upper rear frame 56 and the lower rear frame 55 in the vertical direction, and are formed to have a shorter length in the vehicle front-rear direction than the upper rear frame 56 and the lower rear frame 55.

(40) A first vertical frame 64 extending in the vertical direction is connected between each of the left and right rear sub-frames 63 and the left and right upper rear frames 56. A second vertical frame 65 and a third vertical frame 66 extending in the vertical direction are connected between each of the left and right rear sub-frames 63 and the left and right lower rear frames 55.

(41) A first support frame 67 that extends in the vehicle body front-rear direction and supports the engine 21 is attached to the first and second lower cross frames 57 and 58. A second support frame 68 that extends in the vehicle body front-rear direction and supports the engine 21 is attached to the floor cross frame 53 and the first lower cross frame 57. The second support frame 68 is arranged further on the vehicle body right side than the first support frame 67. A third support frame 69 that extends in the vehicle width direction and supports the engine 21 is attached to the left and right rear sub-frames 63.

(42) As illustrated in FIG. 3, the engine 21 is supported by the vehicle body frame 6 using mount brackets 100 and 200 for mounting the engine 21 on the vehicle body frame 6. The engine 21 is supported by the vehicle body frame 6 in a manner that the first mount bracket 100 arranged on the vehicle body left side and the vehicle body front side of the engine 21, the second mount bracket 200 arranged on the vehicle body right side and the vehicle body front side of the engine 21, and a rear mount 300 provided on the vehicle body rear side of the engine 21 are attached to the first support frame 67, the second support frame 68, and the third support frame 69, respectively. The rear mount 300 is provided on the vehicle body rear side of the transmission 31. The rear mount

300 may be provided on the vehicle body rear side of the engine **21** and attached to the vehicle body frame **6**.

(43) FIG. **5** is a left side view of the power unit, FIG. **6** is a right side view of the power unit, FIG. **7** is a top view of the power unit, and FIG. **8** is a front view of the power unit. FIG. **9** is a front view of the power unit excluding the first mount bracket. As illustrated in FIGS. **5** to **9**, the power unit **20** includes the engine **21** and the transmission **31**, and the engine **21** and the transmission **31** are connected via the connecting bracket **30**.

(44) As illustrated in FIG. **7**, the front wheel propeller shaft **11** is arranged below the engine **21**, and is arranged at a position overlapping the engine **21** in the vehicle front-rear direction and the vehicle width direction in plan view. As illustrated in FIG. **5**, the front wheel propeller shaft **11** is arranged at a position overlapping the engine **21** in the vehicle body vertical direction in a side view (i.e., when viewed from above the vehicle body frame).

(45) As illustrated in FIG. **9**, the crankcase **23** has the recessed portion **41** in which a lower portion of the crankcase **23** is recessed upward and the remaining portion **42** excluding the recessed portion **41** in the axial direction of the crankshaft **22**. At least a part of the front wheel propeller shaft **11** is arranged in a space **43** defined below the recessed portion **41** and is located above a lower end portion of the crankcase **23**.

(46) The oil pan **27** is arranged further on the vehicle body right side than the front wheel propeller shaft **11**, and is formed to be shorter than the crankcase **23** in the vehicle width direction. The oil pan **27** is attached to the remaining portion **42** excluding the recessed portion **41** provided in a lower portion of the crankcase **23** in the vehicle width direction.

(47) As described above, the engine **21** is supported by the vehicle body frame **6** in a manner that the first mount bracket **100**, the second mount bracket **200**, and the rear mount **300** are attached to the first support frame **67**, the second support frame **68**, and the third support frame **69**, respectively.

(48) FIG. **10** is a perspective view illustrating the vicinity of the first mount bracket, FIG. **11** is a perspective view illustrating the vicinity of the first mount bracket excluding the vehicle body frame, FIG. **12** is a cross-sectional view of the vicinity of the first mount bracket taken along line Y12-Y12 in FIG. **11**, and FIG. **13** is a cross-sectional view of the vicinity of the first mount bracket taken along line Y13-Y13 in FIG. **11**.

(49) As illustrated in FIGS. **10** to **13**, the first mount bracket **100** includes a front surface portion **101** arranged on the vehicle body front side and extending in the vehicle width direction, and a side surface portion **102** extending from the vehicle body left side of the front surface portion **101** to the vehicle body rear side. The first mount bracket **100** is formed in a substantially L shape in plan view in a manner that the front surface portion **101** and the side surface portion **102** are integrally formed. As illustrated in FIGS. **5** and **8**, the first mount bracket **100** is arranged at a position overlapping the crankcase **23** in the vehicle width direction, the vehicle body vertical direction, and the vehicle front-rear direction.

(50) The first mount bracket **100** includes a drive source connection portion **110** connected to the engine **21**, a frame connection portion **120** connected to the vehicle body frame **6**, and a related component connection portion **130** connected to the front wheel propeller shaft **11** as a related component related to the engine **21**. The first mount bracket **100** includes, as the drive source connection portion **110**, a front drive source connection portion **111** and a rear drive source connection portion **116** arranged on the vehicle front side and the vehicle body rear side of the first mount bracket **100**, respectively.

(51) As illustrated in FIG. **11**, the front drive source connection portion **111** is provided on the inner side in the vehicle width direction of the front surface portion **101** of the first mount bracket **100**. The front drive source connection portion **111** includes an attachment surface portion **112** formed on the vehicle body rear side of the front surface portion **101** and two bolt insertion holes **113** extending through the attachment surface portion **112** in the front-rear direction. The two bolt

insertion holes **113** are arranged to be offset in the vertical direction and the vehicle width direction.

(52) In a front surface portion **23a** on the vehicle body left side of the crankcase **23**, an attachment surface portion **23b** extending in a direction substantially orthogonal to the front-rear direction is formed corresponding to the two bolt insertion holes **113** of the front drive source connection portion **111**, and two screw holes are formed in the attachment surface portion **23b**. The front drive source connection portion **111** is attached to the crankcase **23** as the attachment surface portion **112** of the front drive source connection portion **111** is placed on the attachment surface portion **23b** of the crankcase **23** and a fastening bolt B1 is screwed into the screw hole through the bolt insertion hole **113**.

(53) The rear drive source connection portion **116** is provided on the vehicle body rear side of the side surface portion **102** of the first mount bracket **100**. The rear drive source connection portion **116** includes an attachment surface portion **117** formed on the inner side in the vehicle width direction of the side surface portion **102**, and two bolt insertion holes extending through the attachment surface portion **117** in the vehicle width direction. The two bolt insertion holes are arranged to be offset in the vertical direction and the vehicle body front-rear direction.

(54) An attachment surface portion **23d** extending in a direction substantially orthogonal to the vehicle width direction is formed on a side surface portion **23c** on the vehicle body left side of the crankcase **23** in correspondence with the two bolt insertion holes of the rear drive source connection portion **116**, and two screw holes are formed in the attachment surface portion **23d**. The rear drive source connection portion **116** is attached to the crankcase **23** as the attachment surface portion **117** of the rear drive source connection portion **116** is placed on the attachment surface portion **23d** of the crankcase **23** and a fastening bolt B2 is screwed into the screw hole through the bolt insertion hole.

(55) The frame connection portion **120** is provided in a portion protruding downward on the center side in the front-rear direction of the side surface portion **102** of the first mount bracket **100**. The frame connection portion **120** includes a cylindrical portion **121** having an axial center extending in the vehicle width direction on the center side in the front-rear direction of the side surface portion **102**, and a rubber bush **122** fitted into the cylindrical portion **121** and formed in a substantially cylindrical shape.

(56) As illustrated in FIG. **13**, the rubber bush **122** is formed in a substantially cylindrical shape, and has flange portions **123** on both sides protruding radially outward as compared with the center side in the axial direction on both sides in the axial direction. The rubber bush **122** is attached in a manner of being fitted into the cylindrical portion **121** with the flange portions **123** on both sides sandwiching the cylindrical portion **121**. A sleeve **124** made from metal formed in a cylindrical shape is joined to the inner side of the rubber bush **122**, and an end surface plate **125** made from metal formed in a disk shape is joined to each of both end surfaces in the axial direction of the rubber bush **122**.

(57) As illustrated in FIG. **4**, the first support frame **67** includes a bottom surface portion **67a** and side surface portions **67b** on both sides, and is formed in a substantially U shape when viewed from the vehicle body front-rear direction. A bush receiving portion **67c** protruding upward corresponding to the end surface plate **125** of the rubber bush **122** is formed in the side surface portion **67b** on the vehicle body right side, and a bolt insertion hole **67d** is formed in the bush receiving portion **67c**. On the side surface portion **67b** on the vehicle body left side, a bush accommodating portion **67e** recessed downward so as to accommodate the end surface plate **125** of the rubber bush **122** is formed, and two bolt insertion holes **67f** are formed in the front-rear direction of the bush accommodating portion **67e**.

(58) As illustrated in FIG. **10**, an attachment plate **126** is formed in a flat plate shape, and has two bolt insertion holes formed on both sides in the vehicle body front-rear direction and one bolt insertion hole formed on the center side in the vehicle body front-rear direction. The attachment

plate **126** is attached to the first support frame **67** with a fastening bolt **B3** and a nut as two bolt insertion holes on both sides are aligned with the two bolt insertion holes **67f** of the first support frame **67**. The nut is provided on the vehicle body right side of the side surface portion **67b** on the vehicle body left side of the first support frame **67**.

(59) The rubber bush **122** is attached to the first support frame **67** as a fastening bolt **B4** is screwed with a nut provided on the vehicle body right side of the bush receiving portion **67c** in a state where one end surface in the axial direction of the rubber bush **122** abuts on the bush receiving portion **67c** and the other end surface in the axial direction of the rubber bush **122** abuts on the attachment plate **126**. The nut is provided on the vehicle body right side of the bush receiving portion **67c**. The first mount bracket **100** is supported by the vehicle body frame **6** with the rubber bush **122** attached to the first support frame **67**.

(60) The related component connection portion **130** is provided on the center side in the vehicle width direction of the front surface portion **101** of the first mount bracket **100**. As illustrated in FIG. **12**, the related component connection portion **130** includes a cylindrical portion **131** whose axial center extends in the vehicle body front-rear direction on the center side in the vehicle width direction of the front surface portion **101**, and a bearing **132** that is fitted into the cylindrical portion **131** and rotatably supports the front wheel propeller shaft **11**.

(61) The bearing **132** includes an outer ring **133** having a substantially cylindrical shape fitted and attached in the cylindrical portion **131**, an inner ring **134** fitted to an outer periphery of the front wheel propeller shaft **11** and concentric with the outer ring **133**, and a plurality of balls **135** interposed between the outer ring **133** and the inner ring **134**, and the outer ring **133** and the inner ring **134** are configured to be relatively rotatable.

(62) As illustrated in FIG. **1**, the front wheel propeller shaft **11** is divided into a first divided shaft **11a** arranged on the vehicle body rear side, a second divided shaft **11b** arranged on the vehicle body front side of the first divided shaft **11a**, and a third divided shaft **11c** arranged on the vehicle body front side of the second divided shaft **11b**. The front wheel propeller shaft **11** includes a rotating portion that rotates by power from the engine **21**, and the first divided shaft **11a**, the second divided shaft **11b**, and the third divided shaft **11c** as the rotating portions rotate by power from the engine **21**.

(63) The first divided shaft **11a** and the second divided shaft **11b** are connected via a first universal joint **11d**, and the second divided shaft **11b** and the third divided shaft **11c** are connected via a second universal joint **11e**. As the first universal joint **11d** and the second universal joint lie, a cruciform shaft joint that is configured such that rotation axes of two members can be bent and rotation can be transmitted is used.

(64) A bevel gear to which power from the engine **21** is transmitted from the front wheel output shaft **33** is attached to a rear end portion of the first divided shaft **11a** of the front wheel propeller shaft **11**, and the first universal joint **11d** is attached to a front end portion of the first divided shaft **11a**. The first divided shaft **11a** is arranged below the engine **21**, and is arranged at a position overlapping the engine **21** in the front-rear direction and the vehicle width direction in plan view and at a position overlapping the engine **21** in the vehicle body vertical direction in side view.

(65) The first divided shaft **11a** is inserted into the cylindrical portion **131** provided in the front surface portion **101** of the first mount bracket **100** and inserted into the bearing **132** to be attached to the first mount bracket **100**. The first divided shaft **11a** is supported by the first mount bracket **100** via the bearing **132**.

(66) In this manner, the first mount bracket **100** is rigidly connected to the engine **21** at the drive source connection portion **110** and is elastically connected to the vehicle body frame **6** at the frame connection portion **120**. The drive source connection portion **110** and the engine **21** are coupled in an abutting state, and the frame connection portion **120** and the vehicle body frame **6** are coupled via the rubber bush **122** which is an elastic member. The first mount bracket **100** is coupled to the front wheel propeller shaft **11** via the bearing **132** at the related component connection portion **130**.

(67) The front drive source connection portion **111**, which is the drive source connection portion **110**, is formed at one end portion in the axial direction of the output shaft **22** of the engine **21** in the first mount bracket **100**. The frame connection portion **120** is formed at the other end portion in the axial direction of the output shaft **22** of the engine **21** in the first mount bracket **100**. The related component connection portion **130** is formed at an intermediate portion between the drive source connection portion **110** and the frame connection portion **120** in the axial direction of the engine **21**.

(68) In the first mount bracket **100**, the drive source connection portion **110** is formed in an upper portion of the first mount bracket **100**, the frame connection portion **120** is formed in a lower portion of the first mount bracket **100**, and the related component connection portion **130** is at least partially formed between the drive source connection portion **110** and the frame connection portion **120** in the vertical direction. As the drive source connection portion **110**, the front drive source connection portion **111** and the rear drive source connection portion **116** are included. However, the configuration may be such that one of the front drive source connection portion **111** and the rear drive source connection portion **116** is included.

(69) FIG. **14** is a perspective view illustrating the vicinity of the second mount bracket, FIG. **15** is a perspective view illustrating the vicinity of the second mount bracket excluding the vehicle body frame, and FIG. **16** is a perspective view illustrating the second mount bracket. As illustrated in FIGS. **14** to **16**, the second mount bracket **200** extends in a substantially flat plate shape in a direction substantially orthogonal to the vehicle width direction and is formed in a substantially trapezoidal shape in side view.

(70) As illustrated in FIG. **15**, the second mount bracket **200** includes a drive source connection portion **210** connected to the engine **21**, a frame connection portion **220** connected to the vehicle body frame **6**, and a related component connection portion **230** connected to an air conditioner compressor **70** as a related component related to the engine **21**.

(71) The drive source connection portion **210** is provided on the vehicle body rear side and the vehicle upper side of the second mount bracket **200**. The drive source connection portion **210** includes an attachment surface portion **211** formed on the vehicle body rear side and the vehicle upper side of the second mount bracket **200**, and two bolt insertion holes **212** extending through the attachment surface portion **211** in the vehicle width direction. The two bolt insertion holes **212** are arranged to be offset in the vertical direction and the vehicle body front-rear direction.

(72) In a side surface portion **23e** on the vehicle body right side of the crankcase **23**, an attachment surface portion **23f** extending in a direction substantially orthogonal to the vehicle width direction is formed corresponding to the two bolt insertion holes **212** of the drive source connection portion **210**, and two screw holes are formed in the attachment surface portion **23f**. The drive source connection portion **210** is attached to the crankcase **23** as the attachment surface portion **211** of the drive source connection portion **210** is placed on the attachment surface portion **23f** of the crankcase **23** and a fastening bolt **B5** is screwed into the screw hole through the bolt insertion hole **212**.

(73) The frame connection portion **220** is provided on the vehicle body lower side of the second mount bracket **200**. As illustrated in FIG. **16**, the frame connection portion **220** includes a cylindrical portion **221** whose axial center extends in the vehicle width direction on the vehicle body lower side of the second mount bracket **200**, and a rubber bush **222** which is fitted in the cylindrical portion **221** and is formed in a substantially cylindrical shape. The rubber bush **222** of the second mount bracket **200** is arranged further on the vehicle body right side and the vehicle body front side than the rubber bush **122** of the first mount bracket **100**.

(74) The rubber bush **222** is formed in the same manner as the rubber bush **122**. The rubber bush **222** is attached in a manner of being fitted into the cylindrical portion **221** with flange portions **223** on both sides sandwiching the cylindrical portion **221**. In the rubber bush **222**, a sleeve **224** is joined to the inner side of the rubber bush **222**, and end surface plates **225** are joined to both end

surfaces in the axial direction of the rubber bush 222.

(75) As illustrated in FIG. 4, the second support frame 68 is formed substantially in the same manner as the first support frame 67. The second support frame 68 includes a bottom surface portion 68a and side surface portions 68b on both sides, and is formed in a substantially U shape when viewed from the vehicle body front-rear direction. A bush receiving portion 68c is formed on the side surface portion 68b on the vehicle body right side, and a bolt insertion hole 68d is formed in a bush receiving portion 68c. A bush accommodating portion 68e and two bolt insertion holes 68f are formed in the side surface portion 68b on the vehicle body left side. As illustrated in FIG. 14, an attachment plate 226 is attached to the second support frame 68 with a fastening bolt B6 and a nut N6.

(76) The rubber bush 222 is attached to the second support frame 68 as a fastening bolt B7 is screwed with a nut in a state where one end surface in the axial direction of the rubber bush 222 abuts on the bush receiving portion 68c and the other end surface in the axial direction of the rubber bush 222 abuts on the attachment plate 226. The nut is provided on the vehicle body right side of the bush receiving portion 68c. The second mount bracket 200 is supported by the vehicle body frame 6 with the rubber bush 222 attached to the second support frame 68.

(77) The related component connection portion 230 is provided on the vehicle front side and the vehicle upper side of the second mount bracket 200. The related component connection portion 230 includes an attachment boss portion 231 extending in the vehicle width direction on the vehicle front side and the vehicle upper side of the second mount bracket 200, and a bolt insertion hole 232 penetrating and extending in the vehicle width direction is formed in the attachment boss portion 231.

(78) The air conditioner compressor 70 as a related component is connected to the engine 21 and arranged on the vehicle body front side and the vehicle body right side of the engine 21. The compressor 70 is arranged at a position overlapping the engine 21 in the vehicle width direction. The compressor 70 is arranged on the vehicle body right side of the engine 21, and the converter 35 is arranged on the vehicle body left side of the engine 21. The compressor 70 is arranged on the opposite side in the axial direction of the output shaft 22 of the engine 21 to the converter 35 with the engine 21 interposed between them.

(79) As illustrated in FIGS. 14 and 15, the compressor 70 includes a compressor case 72 having a substantially cylindrical shape that accommodates a compressor main body portion 71 for compressing a refrigerant, a compressor shaft 73 extending from the inside of the compressor case 72 to the vehicle body right side, an air conditioner pulley 75 to which power from the engine 21 is transmitted from the crankshaft 22 via a belt 74, and a magnet clutch 76 that connects and disconnects power transmission between the compressor shaft 73 and the air conditioner pulley 75.

(80) The air conditioner pulley 75 is attached to the compressor shaft 73 via the magnet clutch 76. The compressor 70 includes the compressor shaft 73 as a rotating portion that rotates by power from the engine 21, and rotates by power from the engine 21.

(81) The compressor 70 includes a lower attachment portion 77 that protrudes radially outward from the lower side of the compressor case 72 and is for attachment to the second mount bracket 200. The compressor 70 includes two of the lower attachment portions 77 provided apart from each other in the axial direction of the compressor case 72. A bolt insertion hole extending in the vehicle width direction is formed in the lower attachment portion 77 on the vehicle body left side, and a screw hole 77a extending in the vehicle width direction is formed in the lower attachment portion 77 on the vehicle body right side.

(82) The compressor 70 includes an upper attachment portion 78 that protrudes radially outward from the upper side of the compressor case 72 and is for attachment to the engine 21. The compressor 70 includes two of the upper attachment portions 78 provided apart from each other in the axial direction of the compressor case 72. A bolt insertion hole extending in the vehicle width direction is formed in the lower attachment portions 77 on the vehicle body left side, and a screw

hole **78a** extending in the vehicle width direction is formed in the lower attachment portions **77** on the vehicle body right side. The upper attachment portion **78** is attached to the engine **21** via an upper attachment bracket **80** different from the second mount bracket **200**.

(83) The compressor **70** is attached to and supported by the second mount bracket **200** as a fastening bolt **B8** is screwed from the vehicle body left side into the screw hole **77a** of the lower attachment portion **77** through the bolt insertion hole of the lower attachment portion **77** and the bolt insertion hole **232** of the attachment boss portion **231** in a state where the attachment boss portion **231** of the second mount bracket **200** is arranged between two of the lower attachment portions **77** of the compressor **70**.

(84) The compressor **70** is also attached to the engine **21** also by the upper attachment bracket **80**. The upper attachment bracket **80** includes a first attachment portion **81** attached to the vehicle body front side of the cylinder block **24** and a second attachment portion **82** attached to the vehicle body right side of the crankcase **23**, and the first attachment portion **81** is provided further on the vehicle body upper side and the vehicle body front side than the second attachment portion **82**. A bolt insertion hole is formed in each of the first attachment portion **81** and the second attachment portion **82**.

(85) The engine **21** is provided with attachment portions **21a** and **21b** on the vehicle body front side and the vehicle body right side of the engine **21** corresponding to the upper attachment bracket **80**, and a screw hole is formed in each of the attachment portions **21a** and **21b**. The first attachment portion **81** and the second attachment portion **82** of the upper attachment bracket **80** are attached to the attachment portions **21a** and **21b** of the engine **21** as the fastening bolts **B8** is screwed into the screw hole through the bolt insertion hole.

(86) The upper attachment bracket **80** includes a third attachment portion **83** mounted to the compressor **70**, and the third attachment portion **83** is provided further on the vehicle body front side than the first attachment portion **81** and the second attachment portion **82**. The third attachment portion **83** includes a cylindrical portion **83a** extending in a substantially cylindrical shape in the vehicle width direction, and a bolt insertion hole is formed in the cylindrical portion **83a**.

(87) The compressor **70** is attached to the upper attachment bracket **80** as the fastening bolt **B8** is screwed from the vehicle body left side into the screw hole **78a** of the upper attachment portion **78** through the bolt insertion hole of the upper attachment portion **78** and the bolt insertion hole of the third attachment portion **83** in a state where the third attachment portion **83** of the upper attachment bracket **80** is arranged between two of the upper attachment portions **78** of the compressor **70**. The compressor **70** may be directly attached to the engine **21** without using the upper attachment bracket **80**. The compressor **70** may be attached only to the second mount bracket **200** without using the upper attachment bracket **80**.

(88) In this manner, the second mount bracket **200** is rigidly connected to the engine **21** at the drive source connection portion **210** and is elastically connected to the vehicle body frame **6** at the frame connection portion **220**. The drive source connection portion **210** and the engine **21** are coupled in an abutting state, and the frame connection portion **220** and the vehicle body frame **6** are coupled via the rubber bush **222** which is an elastic member. The second mount bracket **200** is rigidly connected to the compressor **70** at the related component connection portion **230**. The related component connection portion **230** and the compressor **70** are coupled in an abutting state.

(89) In the second mount bracket **200**, the drive source connection portion **210** is formed in an upper portion of the second mount bracket **200**, the frame connection portion **220** is formed in a lower portion of the second mount bracket **200**, and the related component connection portion **230** is at least partially formed between the drive source connection portion **210** and the frame connection portion **220** in the vertical direction.

(90) In the second mount bracket **200**, the drive source connection portion **210** is formed at a rear portion of the second mount bracket **200**, and the frame connection portion **220** and the related

component connection portion **230** are formed at a front portion of the second mount bracket **200**. The configuration may be such that at least a part of the related component connection portion **230** is formed in an intermediate portion between the drive source connection portion **210** and the frame connection portion **220** in the vehicle body front-rear direction.

(91) FIG. **17** is a perspective view illustrating the vicinity of the rear mount. As illustrated in FIG. **17**, the rear mount **300** is provided integrally with a transmission case **39** on the vehicle body rear side of the transmission **31**. The rear mount **300** is formed separately from the transmission case, and can be attached to the transmission case.

(92) The rear mount **300** includes a cylindrical portion **311** having an axial center extending in the vehicle width direction, and a rubber bush **312** fitted into the cylindrical portion **311** and formed in a substantially cylindrical shape. In the rubber bush **312**, flange portions **313** on both sides are fitted and attached in the cylindrical portion **311** so as to sandwich the cylindrical portion **311**, a sleeve is joined to the inner side of the rubber bush **312**, and an end surface plate **315** is joined to both end surfaces in the axial direction of the rubber bush **312**.

(93) The third support frame **69** is integrally attached to two support plates **69a** spaced apart from each other in the vehicle width direction and extending upward. As illustrated in FIG. **4**, a bush receiving portion **69c** is formed in the support plate **69a** on the vehicle body right side, and a bolt insertion hole **69d** is formed in the bush receiving portion **69c**. A bush accommodating portion **69e** and two bolt insertion holes **69f** are formed in the support plate **69a** on the vehicle body left side. As illustrated in FIG. **17**, an attachment plate **316** is formed in a flat plate shape, and is attached to the support plate **69a** on the vehicle body left side using a fastening bolt **B10** and a nut **N10**.

(94) The rubber bush **312** is attached to the second support frame **68** using a fastening bolt and a nut **N11** in a state where one end surface in the axial direction of the rubber bush **312** abuts on a bush receiving portion **69b** and the other end surface in the axial direction of the rubber bush **312** abuts on the attachment plate **316**. The rear mount **300** is supported by the vehicle body frame **6** as the rubber bush **312** is attached to the third support frame **69**.

(95) In this manner, the engine **21** is elastically connected to the vehicle body frame **6** by the first mount bracket **100**, the second mount bracket **200**, and the rear mount **300** in a state of being connected to the transmission **31**. The first mount bracket **100** and the second mount bracket **200** support the front wheel propeller shaft **11** and the air conditioner compressor **70** as related components connected to the engine **21**, respectively.

(96) The first mount bracket **100** and the second mount bracket **200** are rigidly connected to the engine **21** and are elastically connected to the vehicle body frame **6**. The front wheel propeller shaft **11** is rotatably supported by the first mount bracket **100** via the bearing **132**. As described above, the front wheel propeller shaft **11** is arranged below the crankcase **23**, and at least a part of the front wheel propeller shaft **11** is located further to the upper side than a lower end portion of the crankcase **23** attached to the oil pan **27**.

(97) In the present embodiment, the front wheel propeller shaft **11** and the air conditioner compressor **70** are supported by the first mount bracket **100** and the second mount bracket **200**, but the front wheel propeller shaft **11** may be supported by the first mount bracket **100** while the air conditioner compressor is not supported by the second mount bracket **200**, or the air conditioner compressor **70** may be supported by the second mount bracket **200** while the front wheel propeller shaft is not supported by the first mount bracket **100**.

(98) FIG. **18** is a top view of the crankcase and the oil pan, FIG. **19** is a cross-sectional view of the crankcase and the oil pan taken along line Y19-Y19 in FIG. **18**, and FIG. **20** is a cross-sectional view of the crankcase and the oil pan taken along line Y20-Y20 in FIG. **18**. As illustrated in FIGS. **18** to **20**, an upper mating surface **401** attached to the cylinder block **24** is provided on an upper end surface of the crankcase **23**, and a lower mating surface **402** attached to the oil pan **27** is provided on a lower end surface of the crankcase **23**.

(99) The crankcase **23** has a crank chamber **403** that is recessed downward from the upper mating

surface **401** and accommodates the crankshaft **22** extending in the vehicle width direction. The crank chamber **403** is formed to open to the vehicle body upper side by a front wall portion **404** arranged on the vehicle body front side, a rear wall portion **405** arranged on the vehicle body rear side, a right wall portion **406** arranged on the vehicle body right side, a left wall portion **407** arranged on the vehicle body left side, and a bottom portion **408** arranged on the vehicle body lower side. The crank chamber **403** is partitioned into four crank chambers, specifically, first, second, third, and fourth crank chambers **411**, **412**, **413**, and **414** from the vehicle body left side by three partition wall portions **409** formed between the right wall portion **406** and the left wall portion **407**.

(100) Each of the first, second, third, and fourth crank chambers **411**, **412**, **413**, and **414** accommodates a crank pin of the crankshaft **22** connected to a piston arranged in the cylinder **15** via a connecting rod, a crank arm connected to the crank pin, and a counter weight provided in the crank arm.

(101) In the crankcase **23**, a bearing surface portion **415** recessed downward in a semicircular cross section is formed on each of upper end surfaces of the right wall portion **406**, the left wall portion **407**, and three of the partition wall portions **409**, and the bearing surface portion **415** supports a crank journal of the crankshaft **22**. In the crankcase **23**, a portion constituting the third crank chamber **413** in the front wall portion **404** is provided further on the vehicle body front side than a portion constituting the first, second, and fourth crank chambers **411**, **412**, and **414**.

(102) The crankcase **23** is integrally provided with an oil pan attachment portion **416** protruding further to the lower side than the bottom portion **408** constituting the crank chamber **403**. The oil pan attachment portion **416** is formed in a substantially square cylindrical shape, and a lower end surface of the oil pan attachment portion **416** is the lower mating surface **402** attached to the oil pan **27**. The oil pan attachment portion **416** is provided below the second, third, and fourth crank chambers **412**, **413**, and **414** provided further on the vehicle body right side than the first crank chamber **411**.

(103) The oil pan **27** has a bottom portion **27a** formed in a substantially rectangular shape in plan view and a side wall portion **27b** extending upward from the bottom portion **27a**, and the bottom portion **27a** is formed to bulge downward. In the oil pan **27**, a mating surface **27c** with the crankcase **23** is provided on an upper end surface of the side wall portion **27b**, and an oil storage chamber **27d** for storing oil as lubricating liquid is formed by the bottom portion **27a** and the side wall portion **27b**. The oil pan **27** is arranged below the second, third, and fourth crank chambers **412**, **413**, and **414** provided further on the vehicle body right side than the first crank chamber **411**.

(104) In the bottom portion **408** of the crankcase **23** constituting the second, third, and fourth crank chambers **412**, **413**, and **414**, a discharge hole **417** for discharging oil to an opening **420** opened downward of the crankcase **23** is formed. Oil stored in the oil pan **27** is sucked up by an oil pump, supplied to the crank journal or the like of the crankshaft **22**, flows to the bottom portion **408** of the first, second, third, and fourth crank chambers **411**, **412**, **413**, and **414**, and is returned from the opening **420** to the oil pan **27** through the discharge hole **417**.

(105) As illustrated in FIG. **19**, a communication hole **418** for communication of the first crank chamber **411** and the second crank chamber **412** is provided on the lower side of the partition wall portion **409** dividing the first crank chamber **411** and the second crank chamber **412**. The communication hole **418** provided on the lower side of the partition wall portion **409** is provided over the bottom portion **408** connecting the vehicle body rear side of the front wall portion **404** and the vehicle front side of the rear wall portion **405**.

(106) As illustrated in FIGS. **18** to **20**, the bottom portion **408** of the first crank chamber **411** and the second crank chamber **412** is provided with a passage **419** for guiding oil guided to the first crank chamber **411** to the discharge hole **417** and the opening **420** formed in the bottom portion **408** of the second crank chamber **412** corresponding to the communication hole **418** provided on the lower side of the partition wall portion **409**. The passage **419** is recessed downward in a groove

shape in a substantially semicircular cross section in the bottom portion **408**. As illustrated in FIG. **18**, the passage **419** extends in the vehicle width direction in the axial direction of the crankshaft **22** from the first crank chamber **411** to the second crank chamber **412**. As illustrated in FIG. **20**, the passage **419** is inclined downward from the vehicle body left side toward the vehicle body right side.

(107) In this manner, oil supplied to the crank journal or the like of the crankshaft **22** and flowing to the bottom portion **408** of the first crank chamber **411** flows from the first crank chamber **411** to the second crank chamber **412** through the passage **419** and returns to the oil pan **27** through the discharge hole **417** and the opening **420**.

(108) In the utility vehicle **1**, as illustrated in FIG. **9**, a lower portion of the crankcase **23** in the axial direction of the crankshaft **22** has the recessed portion **41** recessed upward as viewed in the vehicle body front-rear direction and the remaining portion **42** excluding the recessed portion **41**. At least a part of the front wheel propeller shaft **11** is arranged in the space **43** defined below the recessed portion **41** and positioned further to the upper side than a lower end portion of the crankcase **23**.

(109) The crankcase **23** is provided with the oil pan attachment portion **416** in the remaining portion **42**. The crankcase **23** has the opening **420** that opens downward of the crankcase **23** in the remaining portion **42**, and a bottom portion **44** that closes the lower side of the crankcase **23** in the recessed portion **41**. The oil pan **27** is attached to a lower surface of the remaining portion **42**. The bottom portion **44** provided in the recessed portion **41** is constituted by the bottom portion **408** constituting the first crank chamber **411**. As described above, in the crankcase **23**, the passage **419** that guides oil guided to the bottom portion **44** provided in the recessed portion **41** to the discharge hole **417** provided in the bottom portion **408** of the crankcase **23** and the opening **420** of the remaining portion **42** is formed.

(110) The recessed portion **41** is formed at a position corresponding to one of four of the cylinders **15**, that is, the cylinder **15** on the vehicle body left side, and the remaining portion **42** is formed at a position corresponding to the remaining ones of four of the cylinders **15**, that is, three of the cylinders **15** on the vehicle body right side.

(111) As illustrated in FIG. **19**, in the crankcase **23**, in a lower surface **41a** of the recessed portion **41**, a rib portion **45** that connects the lower surface **41a** of the recessed portion **41** and a side surface **41b** on the remaining portion side of the recessed portion **41** is formed. As illustrated in FIG. **18**, the rib portion **45** linearly extends in parallel with the axial direction of the crankshaft **22** with a predetermined width substantially at the center in the vehicle body front-rear direction of the crankcase **23**. The crankcase **23** is reinforced by the rib portion **45**.

(112) Although the engine **21** and the transmission **31** are formed separately, and the engine **21** and the transmission **31** are connected by the connecting bracket **30** to constitute the power unit **20**, a power unit in which the engine and the transmission are integrally formed can also be used.

(113) As described above, the utility vehicle **1** according to the present embodiment includes the vehicle body frame **6**, the traveling drive source **21**, the mount brackets **100** and **200** for mounting the traveling drive source **21** on the vehicle body frame **6**, and the related components **11** and **70** connected to the traveling drive source **21**, and the related components **11** and **70** are supported by the mount brackets **100** and **200**.

(114) By the above, since the related components **11** and **70** are supported by the mount brackets **100** and **200** for mounting the traveling drive source **21** on the vehicle body frame **6**, the number of components can be reduced and a compact configuration can be achieved as compared with a case where the related components **11** and **70** are connected to the traveling drive source **21** using a bracket different from the mount brackets **100** and **200**.

(115) Further, the related components **11** and **70** include the rotating portions **11** and **73** that are rotated by power from the traveling drive source **21**. By the above, the related components **11** and **70** including the rotating portions **11** and **73** rotated by power from the traveling drive source **21** are

supported by the mount brackets **100** and **200** for mounting the traveling drive source **21** on the vehicle body frame **6**, and thus, it is possible to effectively perform power transmission from the traveling drive source **21** to the related components **11** and **70** by shortening a power transmission path for transmitting power from the traveling drive source **21** to the related components **11** and **70**. (116) Further, the mount bracket **100** is rigidly connected to the traveling drive source **21** and is elastically connected to the vehicle body frame **6**, and the related component **11** is supported by the mount bracket **100** via the bearing **132**. By the above, it is possible to suppress transmission of vibration from the traveling drive source **21** to the vehicle body frame **6** by the mount bracket **100**, and it is possible to effectively transmit power from the traveling drive source **21** to the related component **11** by suppressing displacement of the related component **11** with respect to the traveling drive source **21**.

(117) Further, the mount bracket **200** is rigidly connected to the traveling drive source **21** and the related components **70**, and is elastically connected to the vehicle body frame **6**. By the above, it is possible to suppress transmission of vibration from the traveling drive source **21** to the vehicle body frame **6** by the mount bracket **200** and to suppress displacement of the related component **70** with respect to the traveling drive source **21**.

(118) Further, the mount bracket **100** includes the drive source connection portion **111** formed at one end portion in the axial direction of the output shaft **22** of the traveling drive source **21** in the mount bracket **100** and connected to the traveling drive source **21**, the frame connection portion **120** formed at the other end portion in the axial direction of the output shaft **22** of the traveling drive source **21** in the mount bracket **100** and connected to the vehicle body frame **6**, and the related component connection portion **130** formed between the drive source connection portion **111** and the frame connection portion **120** and connected to the related component **11**. By the above, in the mount bracket **100**, since the related component connection portion **130** is formed between the drive source connection portion **111** formed at one end portion and the frame connection portion **120** formed at the other end portion, the related component connection portion **130** is provided at an intermediate portion connecting the drive source connection portion **111** and the frame connection portion **120** of the mount bracket **100** to suppress increase in size of the mount bracket **100**, and the vehicle body frame **6** can compactly support the traveling drive source **21** and the related component **11**.

(119) Further, the mount bracket **100** includes the drive source connection portion **110** formed in an upper portion of the mount bracket **100** and connected to the traveling drive source **21**, the frame connection portion **120** formed in a lower portion of the mount bracket **100** and connected to the vehicle body frame **6**, and the related component connection portion **130** at least partially formed between the drive source connection portion **110** and the frame connection portion **120** in the vertical direction and connected to the related component **11**. By the above, in the mount bracket **100**, since at least a part of the related component connection portion **130** is formed between the drive source connection portion **110** formed at the upper portion and the frame connection portion **120** formed at the lower portion in the vertical direction, the related component connection portion **130** is provided at an intermediate portion connecting the drive source connection portion **110** and the frame connection portion **120** of the mount bracket **100** to suppress increase in size of the mount bracket **100**, and the vehicle body frame **6** can compactly support the traveling drive source **21** and the related component **11**.

(120) Further, the mount bracket **200** includes the drive source connection portion **210** formed at a rear portion of the mount bracket **200** and connected to the traveling drive source **21**, the frame connection portion **220** formed at a front portion of the mount bracket **200** and connected to the vehicle body frame **6**, and the related component connection portion **230** at least partially formed between the drive source connection portion **210** and the frame connection portion **220** in the vehicle body front-rear direction and connected to the related component **70**. By the above, in the mount bracket **200**, at least a part of the related component connection portion **230** is formed

between the drive source connection portion **210** formed at the rear portion and the frame connection portion **220** formed at the front portion in the vehicle body front-rear direction. Therefore, the related component connection portion **230** is provided in an intermediate portion connecting the drive source connection portion **210** and the frame connection portion **220** of the mount bracket **200** to suppress increase in size of the mount bracket **200**, and the vehicle body frame **6** can compactly support the traveling drive source **21** and the related component **70**.

(121) Further, the related component **11** is arranged adjacent to the traveling drive source **21** in a direction orthogonal to the axial direction of the output shaft **22** of the traveling drive source **21**, and is arranged to overlap the traveling drive source **21** at an axial direction position. By the above, the related component **11** arranged adjacent to the traveling drive source **21** in a direction orthogonal to the axial direction of the output shaft **22** of the traveling drive source **21** is arranged so as to overlap the traveling drive source **21** at the axial direction position. Therefore, an amount of protrusion of the related components **11** and **70** from the traveling drive source **21** in the axial direction can be suppressed, and the traveling drive source **21** and the related component **11** can be compactly arranged.

(122) Further, the power transmission mechanism **35** which is connected to the output shaft **22** of the traveling drive source **21** and to which traveling power from the traveling drive source **21** is transmitted is provided, and the related component **70** is arranged on the opposite side in the axial direction of the output shaft **22** of the traveling drive source **21** to the power transmission mechanism **35** with the traveling drive source **21** interposed between them. By the above, it is possible to suppress interference between the related component **70** and the power transmission mechanism **35** connected to the output shaft **22** of the traveling drive source **21** and to which traveling power is transmitted, and to improve the degree of freedom of the layout of the related component **70** supported by the mount bracket.

(123) Further, the mount bracket **100** is connected to a side portion of the traveling drive source **21** on the output side in the axial direction of the output shaft **22** of the traveling drive source **21**. By the above, in a case where power from the traveling drive source **21** is transmitted to the related component **11**, a power transmission path for transmitting the power from the traveling drive source **21** to the related component **11** can be shortened and configured compactly.

(124) Further, the mount brackets **100** and **200** are provided on a side portion of the traveling drive source **21** on both sides in the axial direction of the output shaft **22** of the traveling drive source **21**, and the related components **11** and **70** are supported by at least one of the mount brackets **100** and **200** provided on side portions on both sides of the traveling drive source **21**. By the above, the related components **11** and **70** can be supported using the mount brackets **100** and **200** provided on side portions on both sides in the axial direction of the output shaft **22** of the traveling drive source **21**, and the number of components can be reduced.

(125) Further, the related component **11** is the propeller shaft **11** to which power from the traveling drive source **21** is transmitted. By the above, since the propeller shaft **11** is supported by the mount bracket **100**, the support rigidity of the propeller shaft **11** can be improved while the number of components is reduced as compared with a case of using a bracket that supports the propeller shaft **11** on the vehicle body frame **6** separately from the mount bracket **100**.

(126) Further, the propeller shaft **11** is supported by the mount bracket **100**, the mount bracket **100** has the drive source connection portion **111** connected to the traveling drive source **21** and the frame connection portion **120** connected to the vehicle body frame **6** and extends in a direction orthogonal to the axial direction of the propeller shaft **11**, the drive source connection portion **111** and the frame connection portion **120** are arranged to be separated from each other in a direction orthogonal to the axial direction of the propeller shaft **11**, and the propeller shaft **11** is supported between the drive source connection portion **111** and the frame connection portion **120** of the mount bracket **100** in a direction orthogonal to the axial direction of the propeller shaft **11** via the bearing **132**. By the above, the propeller shaft **11** can be supported by the mount bracket **110** in the

direction orthogonal to the axial direction of the propeller shaft **11**, and the support rigidity of the propeller shaft **11** can be improved.

(127) Further, the traveling drive source **21** has the portion **41** recessed from the outer side to the inner side in a direction orthogonal to the axial direction of the propeller shaft **11**, and the propeller shaft **11** is supported by the mount bracket **100** in the recessed portion **41**. Accordingly, the traveling drive source **21** and the propeller shaft **11** can be arranged in a compact manner.

(128) Further, the related component **70** is the air conditioner compressor **70** driven by the traveling drive source **21**. By the above, since the air conditioner compressor **70** is supported by the mount bracket **200**, the air conditioner compressor **70** can be integrated with the mount bracket **200**, and the air conditioner compressor **70** can be compactly supported.

(129) The utility vehicle **1** according to the present embodiment also includes the engine **21** having the crankcase **23** that supports the crankshaft **22**, and the propeller shaft **11** arranged below the crankcase **23**. The crankcase **23** has the recessed portion **41** in which a lower portion of the crankcase **23** is recessed upward and the remaining portion **42** excluding the recessed portion **41** in the axial direction of the crankshaft **22**. At least a part of the propeller shaft **11** is arranged in the space **43** defined below the recessed portion **41** and positioned further to the upper side than a lower end portion of the crankcase **23**.

(130) By the above, at least a part of the propeller shaft **11** is arranged in the space **43** defined below the recessed portion **41** of the crankcase **23** further on the upper side than a lower end portion of the crankcase **23**, and the recessed portion **41** of the crankcase **23** has a lower surface positioned further on the upper side than the remaining portion **42** of the crankcase **23**.

Accordingly, the propeller shaft **11** and the crankcase **23** can be arranged close to each other while interference between the propeller shaft **11** and the crankcase **23** is prevented. That is, the engine **21** and the propeller shaft **11** can be arranged more compactly than a case where the propeller shaft **11** is arranged on the outer side in the axial direction of the crankcase **23** than the crankcase **23** or a case where the propeller shaft **11** is arranged further on the lower side than a lower end portion of the crankcase **23**.

(131) Further, the engine **21** has the oil pan **27** provided in a lower portion of the crankcase **23**, the crankcase **23** has the opening **420** opened downward of the crankcase **23** in the remaining portion **42** and the bottom portion **44** closing the lower side of the crankcase **23** in the recessed portion **41**, and the oil pan **27** is attached to a lower surface of the remaining portion **42**. By the above, since the oil pan **27** is attached to the lower surface of the remaining portion **42** of the crankcase **23** having the opening **420** opened downward of the crankcase **23**, it is possible to prevent interference between the oil pan **27** and the propeller shaft **11** arranged in the space **43** defined below the recessed portion **41** of the crankcase **23**.

(132) Further, in the crankcase **23**, the passage **419** for guiding lubricant guided to the bottom portion **44** of the recessed portion **41** to the opening **420** of the remaining portion **42** is formed. By the above, even in a case where the oil pan is not provided below the recessed portion **41** of the crankcase **23**, the lubricant guided to the bottom portion **44** of the recessed portion **41** of the crankcase **23** through the passage **419** can be returned to the oil pan **27**.

(133) Further, the engine **21** is the engine **21** in which a plurality of the cylinders **15** are arranged side by side in the vehicle width direction, the recessed portion **41** is formed at a position corresponding to one of a plurality of the cylinders **15**, and the remaining portion **42** is formed at a position corresponding to a remaining one of a plurality of the cylinders **15**. By the above, since the engine **21** is the engine **21** in which a plurality of the cylinders **15** are arranged side by side in the vehicle width direction, the utility vehicle **1** including a transversal type multiple cylinder engine can be configured compactly in the vehicle width direction and the vertical direction.

(134) Further, the rib portion **45** that connects the lower surface **41a** of the recessed portion **41** and the side surface **41b** on the remaining portion side of the recessed portion **41** is formed on the lower surface **41a** of the recessed portion **41**. By the above, the recessed portion **41** of the crankcase **23**

can be reinforced by the rib portion **45**.

(135) In the present embodiment, the front wheel propeller shaft **11** is arranged at a position corresponding to one of the cylinders **15** arranged on the vehicle body left side of the crankcase **23**, but can be arranged at a position corresponding to one of the cylinders **15** arranged on the vehicle body right side of the crankcase **23**. Although the engine **21** is used as the traveling drive source, a motor can also be used as the traveling drive source.

(136) The present disclosure is not limited to the illustrated embodiment, and various improvements and design changes can be made without departing from the gist of the present disclosure.

Claims

1. A utility vehicle comprising: a vehicle body frame; a traveling drive source; a mount bracket for mounting the traveling drive source on the vehicle body frame; and a propeller shaft connected to the traveling drive source, and extending in a vehicle body front-rear direction, the propeller shaft being configured to transmit power from the traveling drive source to a front wheel, wherein at least a portion of the propeller shaft is located at a position overlapping the traveling drive source when viewed from a position above the vehicle body frame, wherein the propeller shaft extends forward from behind the traveling drive source and is supported by the mount bracket at a position forward of the position overlapping the traveling drive source when viewed from the position above the vehicle body frame, and wherein: the mount bracket is rigidly connected to the traveling drive source and is elastically connected to the vehicle body frame, and the propeller shaft is rotatably supported by the mount bracket via a bearing.
2. The utility vehicle according to claim 1, wherein the mount bracket includes: a drive source connection portion at a first axial end of an output shaft of the traveling drive source in the mount bracket and connected to the traveling drive source, a frame connection portion at a second axial end of the output shaft of the traveling drive source in the mount bracket and connected to the vehicle body frame, and a related component connection portion formed between the drive source connection portion and the frame connection portion and connected to the propeller shaft.
3. The utility vehicle according to claim 1, wherein the mount bracket includes: a drive source connection portion in an upper portion of the mount bracket and connected to the traveling drive source, a frame connection portion in a lower portion of the mount bracket and connected to the vehicle body frame, and a related component connection portion at least partially between the drive source connection portion and the frame connection portion in a vertical direction and connected to the propeller shaft.
4. The utility vehicle according to claim 1, wherein the propeller shaft is arranged adjacent to the traveling drive source in a direction orthogonal to an axial direction of an output shaft of the traveling drive source, and the related component is arranged to overlap the traveling drive source at an axial direction position.
5. The utility vehicle according to claim 1, wherein the mount bracket is connected to a side portion of the traveling drive source on an output side in an axial direction of an output shaft of the traveling drive source.
6. The utility vehicle according to claim 1, wherein: the mount bracket includes a drive source connection portion connected to the traveling drive source and a frame connection portion connected to the vehicle body frame, and the mount body extends in a direction orthogonal to an axial direction of the propeller shaft, the drive source connection portion and the frame connection portion are arranged apart from each other in a direction orthogonal to the axial direction of the propeller shaft, and the propeller shaft is supported between the drive source connection portion and the frame connection portion of the mount bracket via a bearing in a direction orthogonal to the axial direction of the propeller shaft.

7. The utility vehicle according to claim 1, wherein: the traveling drive source has a portion recessed from an outer side to an inner side in a direction orthogonal to an axial direction of the propeller shaft, and the propeller shaft is supported by the mount bracket in the recessed portion.
8. The utility vehicle according to claim 1, further comprising: a transmission including an input shaft to which power from the traveling drive source is transmitted, a front wheel output shaft for transmitting power from the traveling drive source to the front wheel, and a rear wheel output shaft for transmitting power from the traveling drive source to a rear wheel, wherein the propeller shaft is configured to transmit the power transmitted to the input shaft to the front wheel.
9. The utility vehicle according to claim 8, wherein: the transmission further includes a two-wheel drive four-wheel drive switching device, and the two-wheel drive four-wheel drive switching device is configured to be switchable between a two wheel drive state in which the input shaft and the rear wheel output shaft are connected and a four wheel drive state in which the input shaft, the front wheel output shaft, and the rear wheel output shaft are connected.
10. A utility vehicle comprising: an engine having a crankcase supporting a crankshaft; a propeller shaft arranged below the crankcase, and a mount bracket for mounting the engine on a vehicle body frame, wherein the crankcase includes a recessed portion in which a lower portion of the crankcase is recessed upward and a remaining portion excluding the recessed portion in an axial direction of the crankshaft, wherein at least a part of the propeller shaft is arranged in a space defined below the recessed portion, and positioned further on an upper side than a lower end portion of the crankcase, wherein the propeller shaft is supported by the mount bracket, wherein the engine includes a plurality of cylinders arranged side by side in a vehicle width direction, wherein the recessed portion is formed at a position corresponding to one of the plurality of cylinders, and wherein the remaining portion is formed at a position corresponding to a remaining one of the plurality of cylinders.
11. The utility vehicle according to claim 10, wherein: the engine includes an oil pan in a lower portion of the crankcase, the crankcase has an opening opened downward of the crankcase in the remaining portion and a bottom portion closing a lower side of the crankcase in the recessed portion, and the oil pan is attached to a lower surface of the remaining portion.
12. The utility vehicle according to claim 11, wherein the crankcase has a passage for guiding lubricant guided to a bottom portion of the recessed portion to the opening of the remaining portion.
13. The utility vehicle according to claim 10, wherein: the mount bracket is rigidly connected to the engine and is elastically connected to the vehicle body frame, and the propeller shaft is rotatably supported by the mount bracket via a bearing.
14. The utility vehicle according to claim 10, wherein the mount bracket is connected to a side portion of the engine on an output side in an axial direction of an output shaft of the engine.
15. A utility vehicle comprising: an engine having a crankcase supporting a crankshaft; a propeller shaft arranged below the crankcase, and a mount bracket for mounting the engine on a vehicle body frame, wherein the crankcase includes a recessed portion in which a lower portion of the crankcase is recessed upward and a remaining portion excluding the recessed portion in an axial direction of the crankshaft, wherein at least a part of the propeller shaft is arranged in a space defined below the recessed portion, and positioned further on an upper side than a lower end portion of the crankcase, wherein the propeller shaft is supported by the mount bracket, and wherein a rib portion connecting a lower surface of the recessed portion and a side surface on the remaining portion side of the recessed portion is formed on the lower surface of the recessed portion.
16. The utility vehicle according to claim 15, wherein: the engine includes an oil pan in a lower portion of the crankcase, the crankcase has an opening opened downward of the crankcase in the remaining portion and a bottom portion closing a lower side of the crankcase in the recessed portion, and the oil pan is attached to a lower surface of the remaining portion.
17. The utility vehicle according to claim 16, wherein the crankcase has a passage for guiding

lubricant guided to a bottom portion of the recessed portion to the opening of the remaining portion.

18. The utility vehicle according to claim 15, wherein: the mount bracket is rigidly connected to the engine and is elastically connected to the vehicle body frame, and the propeller shaft is rotatably supported by the mount bracket via a bearing.

19. The utility vehicle according to claim 15, wherein the mount bracket is connected to a side portion of the engine on an output side in an axial direction of an output shaft of the engine.
